

COAL



MORE THAN two hundred million years ago huge trees grew in the warm, humid swamps that covered vast regions of the world. They captured the Sun's energy to make their wood. When they died, their trunks became buried and gradually changed into coal. When we burn coal today, we release the energy the trees captured all those years ago. Because of its origin, coal is called a fossil fuel. It was the first fossil fuel to be used by people, and is still today second only to petroleum in importance for energy production worldwide.

Coal mining

Almost 5 billion tonnes of coal are mined a year. China and the United States mine the most coal, with annual outputs exceeding 1.6 billion tonnes. Coal deposits can be up to about 20 m (66 ft) thick, but they average less than 3 m (10 ft). Some deposits are found on the surface, but most lie underground, sandwiched between rock layers.



Collecting coal at a strip mine

Strip mining

One method of surface, or opencast, mining is called strip mining. The coal is excavated in a series of long strips. Any soil above each strip is used to fill in the trench created when the coal has been removed from a previous strip.



Drilling coal in a shaft mine

Shaft mining

Coal seams deep below the surface are reached by a system of vertical shafts and horizontal tunnels. The coal is dug out by powered coal-cutters and hydraulic tools.



Mine safety
Mines are dangerous places because of the risk of rock falls and the build-up of explosive gases, such as methane. One safety device was invented by an English scientist, Humphry Davy, in 1815. His safety lamp was able to detect dangerous levels of poison gases.

Davy lamp

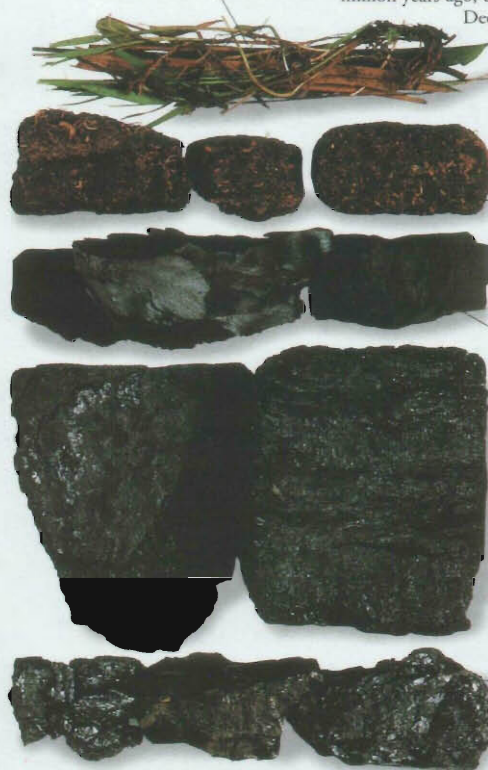
Power

About 25 per cent of the world's energy supply is generated from coal. In coal-fired power stations, the coal is first pulverized (powdered) and then burned in a furnace. The hot gases produced pass over tubes containing water and turn it into steam. The steam drives powerful turbogenerators, which produce electricity. The electricity is then transmitted through a national grid network.



Coal-fired power station, Germany

Piles of dead plant material accumulate in swampy regions



How coal is formed

Coal began to form in swampy forests about 350 million years ago, during the Carboniferous period. Decaying plants were buried under layers of mud. As heat and pressure increased, plant remains slowly converted into coal. Today, there are three main grades of coal – lignite, bituminous coal, and anthracite.

Peat represents an early stage in coal-formation. It is soft, fibrous, and moist, but still gives off heat when burned.

Lignite, or brown coal, is a low-grade fuel containing up to about 60 per cent carbon, along with plant remains and moisture. It is soft and crumbly.

Bituminous coal is a better quality fuel, comprising more than 80 per cent carbon. It is the most common solid fuel used in industry. It is hard, but dirty to the touch.

Anthracite is the highest grade coal, containing more than 90 per cent carbon. It is shiny black, clean to touch, and burns with little smoke.

Coal products

Coal can be processed into valuable products, by a method called destructive distillation. Coal is heated in coke ovens at up to 1,300 °C (2,400 °F) without air. A mixture of liquid vapours and gases escapes and is then separated into coal gas, ammonia liquid, and coal tar. The solid left behind is called coke.



Coke

This solid, porous substance is, like coal, an excellent fuel, which contains more than 80 per cent carbon. It is widely used in industry, mostly in blast furnaces for making iron. In the furnace, it also acts as a chemical agent in the iron-extraction reaction.



Coal tar soap

Coal tar

Coal tar is a black oily liquid that is a rich source of mostly organic chemicals, such as benzene, phenol, and creosote. These can be processed into a variety of materials including dyes, paints, and drugs.

Domestic fuel

Until the mid 1900s, coal was the fuel most used in Western homes. Each room was heated by open coal fires, and cooking was usually done on a coal-burning stove. Today, few modern homes have coal fires, as people tend to use other forms of heating. Some cities and towns allow only smokeless fuels to be used for energy.

Burning smokeless fuels keeps pollution low



FIND OUT MORE

CHEMISTRY

DYES AND PAINTS

ELECTRICITY

ENERGY

FIRE

INDUSTRIAL REVOLUTION

IRON AND STEEL

OIL

PLASTICS AND RUBBER